

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Computer Science and Engineering
2. **Subject Name:** Database Management Systems Laboratory L-T-P: 0-0-3 Credits: 2
3. **Pre-requisites:** Discrete Structures (CS21201)
4. **Syllabus and reference books:**

Syllabus: Database schema design, database creation, SQL programming and report generation using a commercial RDBMS like ORACLE/SYBASE/DB2/SQL-Server/INFORMIX. Students are to be exposed to front end development tools, ODBC and CORBA calls from application Programs, internet based access to databases and database administration.

Reference Books:

- 1) Abraham Silberschatz, Henry Korth, and S. Sudarshan, Database System Concepts, McGrawHill.
- 2) Raghu Ramakrishnan, Database Management Systems, WCB/McGraw-Hill.
- 3) Bipin Desai, An Introduction to Database Systems, Galgotia.
- 4) J. D. Ullman, Principles of Database Systems, Galgotia.
- 5) R. Elmasri and S. Navathe, Fundamentals of Database Systems, Addison-Wesley.
- 6) Serge Abiteboul, Richard Hull and Victor Vianu, Foundations of Databases. Addison-Wesley.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Entity Relationship Model and Relational Database Design	2
2.	Structured Query Language (SQL)	3
3.	Advanced SQL	1
4.	Data Warehousing	1
5.	Web Application Development	4
6.	Smartphone Application development	2
Total number of hours		39 (13x3hrs)